Cypher Stack

Lecture 1 - An Introduction to Cryptography

What we need to understand it

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Cypher Stack Academy | October 28, 2026



Outline



1. The Problem

2. Preliminary Ideas

3. Modular Arithmetic

Section 1

The Problem

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If Alfred and Beatrice are able to give each other a key before hand, they would be able to communicate safely using **symmetric key cryptography**.

This is (more or less) a solved problem. Alfred and Beatrice would have a large number of methods to ensure their communication stays private, with any security guarantees they wish.

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Since this is generally the case, our focus will be to figure out how it might be possible for two people (who potentially have never met) to communicate securely without having any prior setup.

Section 2

Preliminary Ideas

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In order to make the most out of this, I would recommend keeping a small notebook apart from your course notes where you can write down *notation* and *theorems*. This can serve as a sort of 'translation guide' until you get comfortable with the symbols.

Preliminary Ideas

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$\{a \in S : f(a)\}$	All elements of S which have property f	Set builder notation

Practice Problems

Set Theory



Let's look at the first problem set in the worksheet.

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$\mathbb{C}$	The set of complex numbers	$\{a + bi : a, b \in \mathbb{R}\}$

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We will use these quantifiers in order to describe the properties that certain sets. One thing that is very important to keep in mind is that statements can have multiple quantifiers, and the order is **very** important.

Practice Problems

Quantifiers



Let's look at the second set of problems on the worksheet.

Preliminary Ideas

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Definition 1

Let S and T be sets. A function f is an object which takes an element from S and 'sends' it to T. We denote this as:

$$f : S \to T$$

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As a bit of a refresher, let's look at the special properties that functions can possess.

Function Properties

Injective



Definition 2

A function $f : S \to T$ is called **injective** (or one to one) if

$$f(a) = f(b) \Rightarrow a = b$$

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Injectivity is an important property since it guarantees that the outputs of two different inputs will **never** be the same.

Function Properties

Surjective



Definition 3

A function $f : S \to T$ is called **surjective** (or onto) if

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Surjectivity is useful cryptographically by maximizing our key space. It is an advantageous property to have when proving various security properties.

Function Properties

Bijective



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A function $f : S \to T$ is called **bijective** if it is both injective and surjective.

Function Properties

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This is the strongest of the three properties and very useful for cryptographic applications. We can think of it as guaranteeing that every ciphertext can be decoded **and** decoded uniquely.

Practice Problems

Functions



Now let's take a look at some problems involving functions.

Section 3

Modular Arithmetic

Modular Arithmetic

How It Works



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Modular Arithmetic

How It Works



We are going to end off this lecture with a new type of arithmetic: **modular arithmetic**.

The easiest way to think about it is like doing arithmetic on a clock. Whenever we get to 12, instead of counting forward to 13, we wrap back around to 1. However, with modular arithmetic, we always wrap around to 0 and we can choose which number makes the wrap around happen.

Modular Arithmetic

Definitions



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Let's count up to 13 normally, and at the same time, we will count with the wrap around:

Normal	1	2	3	4	5	6	7	8	9	10	11	12	13
Wrapping	1	2	3	4	0	1	2	3	4	0	1	2	3

Modular Arithmetic

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Whenever two values are the same in the same modulus, we say that they are **congruent modulo** n. We represent this by the following symbol:

$$13 \equiv 3 \pmod 5$$

Modular Arithmetic

Operations Continued



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Operations Continued



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The one thing we need to be very careful about is **zero**. In ordinary arithmetic, we aren't allowed to divide by zero and the same rule applies here. **However**, zero can take on a lot of different forms when we introduce modular arithmetic.

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Operations Continued



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In the example prior, when working modulo 5, every multiple of 5 is going to behave like zero! So, we need to be extra careful not to accidentally commit one of the cardinal sins of mathematics: Dividing by zero.